

Lakhdar Berache

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RESEARCH INTERESTS

Autonomous and multi-agent systems built on large language models: agent orchestration and planning, evidence grounding and provenance, safety envelopes and human-in-the-loop supervision, inference under embedded and offline constraints, and the evaluation methodology these systems require. I want to contribute to work that advances the state of the art and proves useful to the people who rely on it, wherever an automated decision carries real consequences: industrial operations, robotics and vehicle software.

EDUCATION

University of Technology of Troyes (UTT) Troyes, France
Master of Engineering (Diplôme d'Ingénieur), Computer Science & Information Systems 2022 – 2027

- Specialisation in Informatics and Systems (ISI); integrated preparatory program 2022–2024.

Korea Advanced Institute of Science and Technology (KAIST) Daejeon, South Korea
Exchange semester; Informatics and Artificial Intelligence (taught in English) Feb – Jun 2025

- Coursework in artificial intelligence, machine learning, reinforcement learning, data science and startup management.
- Team research projects in applied machine learning and wearable sensing, taken from problem framing through prototype to evaluation.

RESEARCH & ENGINEERING EXPERIENCE

D-Valentin Paris, France (remote)
Co-Founder & AI Lead Jul 2026 – Present

- Lead software and AI architecture: orchestration framework, model selection and evaluation methodology.
- Designed the multi-agent pipeline that turns unstructured company material into individual claims, each carrying its source, date and confidence, so that a downstream decision can be audited rather than trusted.
- Venture born at Google's Build with Gemini XPRIZE; team of engineers, investors and operators across Europe.

ArcherMind Technology (Nanjing) Co., Ltd. Paris, France (remote)
Software & AI Engineer, freelance Mar 2026 – Present

- Co-developed Renault Korea's AI Orchestrator, the in-vehicle AI system presented on stage at NextRise 2026 as the centrepiece of Renault Korea's mobility strategy, alongside Kakao Mobility, TMAP Mobility and Valeo.
- Design and evaluate multi-agent workflows (LangGraph, LangChain) for in-car intent understanding, context handling and vehicle-function orchestration through tool-calling agents.
- Tune agent pipelines for graceful degradation rather than best-case behaviour, under embedded constraints: limited compute, offline tolerance, automotive HMI and functional-safety requirements.
- Led one project end to end, from technical design and stack selection through integration to production rollout.

Renault Group – Renault Korea R&D Yongin, South Korea
Software & AI Engineer Intern Sep 2025 – Feb 2026

- Initiated the AI Orchestrator that Renault later showcased at NextRise 2026, and still contribute to it today.
- Prototyped and integrated AI features for infotainment and connected services: personalisation, in-car assistant, generated in-car content.
- Built a multi-agent conversational system with LangGraph, integrating MCP servers, a RAG pipeline over vehicle-manual knowledge, real-time TTS streaming and cloud messaging to mobile.
- Took features from prototype to tested components across connected services and embedded targets; presented monthly technical progress in English to director- and executive-level stakeholders.

SELECTED PROJECTS

Astyr *astyr.app* 2026
Autonomous diagnosis and process optimization for industrial operations | main project

- **Problem.** The fault-diagnosis reflexes of a plant's most experienced maintenance engineers are not written down anywhere, and that generation is retiring.
- **Method.** Documentation, sensor telemetry, maintenance records and past incidents are structured into a living technical model, against which diagnosis and optimisation workflows run. The autonomy is bounded on purpose: every output

- traces back to operational evidence, the system asks instead of guessing below a confidence threshold, and the technician's field verdict closes the loop.
- **Result.** In production across automotive, marine and infrastructure assets. Started as ARC at the RAISE Summit Hackathon Paris, 2nd place in the Vultr track.

Wallside

2026

How retrieval accuracy degrades as a document corpus grows

- **Problem.** An assistant answering from car owner manuals must identify the right manual before the right page. Across 165 manuals, 62,712 pages and 138,087 passages, near-duplicate documents make that the hard part.
- **Method.** Evaluation harness built on the production indexes: questions written by a model generation distinct from the system under test and kept only when their evidence span really occurs in the passage, eight query forms each, nested corpus sizes from 1 to 165 with random and same-brand distractors, dev/test split and 95% confidence intervals.
- **Result.** Document identification falls from 100% on a single manual to 17–27% at 165 for every standard retriever, lexical, dense and hybrid alike. Metadata routing over a weighted hybrid restores 98.4% top-1 at full corpus size, and 92.8% passage hit rate at rank 5 once a cross-encoder reranks.

Alpin *alpin.aminssutt.fr*

2026

Autonomous multi-agent verification of an embedded control system

- **Problem.** Verification of an automotive ECU relies on hand-written scenarios, so rare failure paths through the state machine are systematically under-tested; the space is far too large to enumerate by hand.
- **Method.** An autonomous multi-agent system explores the state machine adversarially, generating its own scenarios, injecting faults and checking safety properties, with a live feedback loop that steers the next round of exploration towards the uncovered regions.
- **Result.** Coverage gaps surfaced and closed iteratively without hand-authored test cases.

Blaze *blaze-live.aminssutt.fr*

2026

Autonomous perception-to-action loop under human command, fully offline

- **Problem.** Fireground radio is dense, unstructured and real time; commanders lose critical minutes structuring it by hand. France recorded 12,500 fire starts and 44,000 ha burned in 2026.
- **Method and result.** Eight-stage pipeline from streaming speech-to-text to synthesised voice order, with fact extraction constrained to quote the transcript, eight hard-coded safety rules acting as a hard envelope and mandatory commander approval. Local Gemma agents on a single machine, zero cloud calls, so the loop holds with no network. Complete voice-to-voice loop running live; built at the Gemma 4 Hackathon Paris.

AWARDS & DISTINCTIONS

- **2nd place, Vultr track,** RAISE Summit Hackathon Paris, Jul 2026 – built ARC in under 24 hours among 6,000+ builders from 60 countries.
- **Public's Favourite Award & Finalist,** Megawatt'UTT Challenge 26 (EDF / Fondation UTT), Jun 2026 – SimuPont, a portable training simulator for the nuclear industry, pitched to an audience of 500 and a jury of industry professionals; covered in *Le Parisien*.

SKILLS, LANGUAGES & SERVICE

Languages	French (native), Arabic (bilingual), English (fluent), Korean (beginner).
Programming	Python, TypeScript / JavaScript, Java, SQL.
AI & ML	LLM agents and multi-agent orchestration (LangGraph, LangChain, MCP); autonomous planning and tool-calling agents; retrieval-augmented generation (hybrid retrieval, metadata routing, reranking, evaluation with confidence intervals); on-device inference (Gemma, vLLM); computer vision; NLP; Hugging Face Transformers, scikit-learn, FAISS, BM25.
Systems	ROS2; embedded and offline deployment; digital twins; industrial and automotive HMI; functional-safety constraints; Docker, FastAPI, Flask, React, PostgreSQL, MongoDB, Supabase, AWS (Bedrock), Git.
Certifications	Getting Started with AWS Generative AI for Developers (AWS, 2025); Build RAG Applications (IBM, 2025).
Service	Ambassador, Hackathon Hub (2026 – present), a European hackathon network of 4,000+ members.